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Effects of Semaglutide on Heart Failure Outcomes in Diabetes and Chronic Kidney Disease in the FLOW Trial

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ABSTRACT

BACKGROUND People with type 2 diabetes (T2D) and chronic kidney disease (CKD) are at high risk for heart failure (HF) and premature death from cardiovascular (CV) causes. The FLOW (Research Study To See How Semaglutide Works Compared to Placebo in People With Type 2 Diabetes and Chronic Kidney Disease), which enrolled participants with T2D and CKD, demonstrated that semaglutide, a glucagon-like peptide-1 receptor agonist, reduced the incidence of the primary composite outcome (persistent ≥50% decline in estimated glomerular filtration rate, persistent estimated glomerular filtration rate <15 mL/min/1.73 m^2, kidney replacement therapy, and kidney or CV death) by 24%.

OBJECTIVES This prespecified analysis examined the effects of semaglutide on HF outcomes in this high-risk population.

METHODS Participants were randomized (1:1) to once-weekly subcutaneous semaglutide 1 mg or placebo. The prespecified main outcome was a composite of HF events (new onset or worsening of HF leading to an unscheduled hospital admission or an urgent visit, with initiation of or intensified diuretic/vasoactive therapy) or CV death. HF data were collected by the investigator. CV death was adjudicated by an independent committee.

RESULTS A total of 3,533 randomized participants were followed for a median of 3.4 years. HF was present at baseline in 342 participants (19.4%) in the semaglutide group and 336 (19.0%) in the placebo group. In the overall trial population, semaglutide increased time to first HF events or CV death (HR: 0.73; 95% CI: 0.62-0.87; $P = 0.0005$), HF events alone (HR: 0.73; 95% CI: 0.58-0.92; $P = 0.0068$), and CV death alone (HR: 0.71; 95% CI: 0.56-0.89; $P = 0.0036$). The risk reduction for the composite HF outcome was similar in those with (HR: 0.73; 95% CI: 0.54-0.98; $P = 0.0338$) and without (HR: 0.72; 95% CI: 0.58-0.89; $P = 0.0028$) HF at baseline. The risk of HF outcomes (HF events or CV death) was generally higher in participants categorized as NYHA functional class III and those with the HF reduced ejection fraction subtype, regardless of treatment.

CONCLUSIONS Semaglutide substantially reduced the risk of time to first composite outcome of HF events or CV death, as well as HF events and CV death alone, in a high-risk population with T2D and CKD. These effects were consistent regardless of history of HF. (A Research Study To See How Semaglutide Works Compared to Placebo in People With Type 2 Diabetes and Chronic Kidney Disease [FLOW]; NCT03819153) (JACC. 2024;84:1615–1628)
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ABBREVIATIONS AND ACRONYMS

ASCVD = atherosclerotic cardiovascular disease

CKD = chronic kidney disease

CV = cardiovascular

eGFR = estimated glomerular filtration rate

GLP-1 RA = glucagon-like peptide-1 receptor agonist

HF = heart failure

HFpEF = heart failure with preserved ejection fraction

HFrEF = heart failure with reduced ejection fraction

IV = intravenous

KDIGO = Kidney Disease: Improving Global Outcomes

LVEF = left ventricular ejection fraction

MRA = mineralocorticoid receptor antagonist

NNTs = numbers needed to treat

PY = patient-years

SGLT2 = sodium-glucose cotransporter 2

T2D = type 2 diabetes

UACR = urine albumin-to-creatinine ratio

Heart failure (HF) affects an estimated 64 million people worldwide.[1] The increasing global prevalence of HF is due to an aging population, as well as increasing rates of hypertension, coronary artery disease, obesity, type 2 diabetes (T2D), and chronic kidney disease (CKD).[2-8] When T2D and CKD coexist, the risk for HF and mortality is increased beyond that of either condition alone.[9] Therefore, developing better strategies for HF prevention and treatment in high-risk patients with T2D is an important clinical priority.

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Despite effective therapies for CKD, the residual risk for CKD progression and especially HF events remains high as evidenced by the demonstrably high rates of HF in the CREDENCE (Evaluation of the Effects of Canagliflozin on Renal and Cardiovascular Outcomes in Participants With Diabetic Nephropathy) study, which examined the efficacy of the sodium-glucose cotransporter 2 (SGLT2) inhibitor canagliflozin in participants with T2D and CKD.[10,11] Emerging evidence suggests a possible benefit of treatment with glucagon-like peptide-1 receptor agonists (GLP-1 RAs) in patients with T2D with CKD and HF.[12-14] FLOW (Research Study To See How Semaglutide Works Compared to Placebo in People With Type 2 Diabetes and Chronic Kidney Disease), the first dedicated trial that examined the effects of a GLP-1 RA on kidney outcomes in participants with T2D and CKD,[15] demonstrated that once-weekly subcutaneous semaglutide 1.0 mg decreased the primary composite outcome of kidney failure, ≥50% persistent estimated glomerular filtration rate (eGFR) decline, or kidney or cardiovascular (CV) death by 24%.[12] In this prespecified analysis, we report the effects of semaglutide on HF outcomes among participants in the FLOW trial.[12]

METHODS

DESIGN. The design of the phase 3 double-blind, randomized, placebo-controlled, international FLOW trial has been described previously.[15] The trial was designed and overseen by a Steering Committee in partnership with members of the study sponsor, Novo Nordisk, who also managed trial operations. The Steering Committee provided academic and scientific leadership for the conduct and analysis of the trial and was responsible for reporting the results. An external, independent adjudication committee was responsible for the adjudication of predefined events. The trial was conducted in accordance with Good Clinical Practice guidelines and the Declaration of Helsinki. Ethical approvals were obtained from relevant regulatory authorities and Institutional Review Boards, and each participant provided written informed consent prior to any trial-related activity.

Technical editorial assistance was provided by OPEN Health Communications and funded by the study sponsor Novo Nordisk. Analyses were conducted by the sponsor and were independently verified with the use of the original data by Statogen Consulting.

PARTICIPANTS. Full inclusion and exclusion criteria have been published previously.[15] In summary, the trial enrolled adults with preexisting T2D and CKD, the latter defined as an eGFR of 50 to 75 mL/min/1.73 m^2 and urine albumin-to-creatinine ratio (UACR) of >300 to <5,000 mg/g (eGFR ≥50 mL/min/1.73 m^2), or a UACR of >100 to <5,000 mg/g if eGFR was 25 to <50 mL/min/1.73 m^2. Participants were undergoing treatment with a stable maximum-labeled or tolerated dose of an angiotensin-converting enzyme inhibitor or angiotensin II receptor blocker and could also be treated with an SGLT2 inhibitor or mineralocorticoid receptor antagonist (MRA). The presence or absence of HF was recorded for all participants at baseline. Individuals with HF were eligible for inclusion into the trial, but those with NYHA functional class IV HF were excluded. Echocardiography and collection of natriuretic peptide levels were not performed as part of the study procedures. Information on left ventricular ejection fraction (LVEF) was not required but was recorded when known.

TRIAL PROCEDURES. Eligible participants were randomized, in a 1:1 ratio, to receive once-weekly subcutaneous injections of semaglutide 1.0 mg or

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The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the Author Center.

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matching placebo. The investigational product was titrated (as tolerated) over a period of 8 weeks (0.25 mg for 4 weeks and 0.5 mg for 4 weeks), followed by a maintenance dose of 1.0 mg for the remainder of the treatment period. Randomization was stratified by SGLT2 inhibitor use at baseline. The concomitant use of glucose-lowering (except GLP-1 RA), CKD, and CV disease standard-of-care medications was permitted throughout the trial in accordance with local guidelines.

STUDY OUTCOMES. CV events were adjudicated by an independent adjudication committee in accordance with prespecified criteria. Members of the adjudication committee were blinded to trial group assignment. HF events, with the exception of CV death, were not adjudicated. During the trial, investigators captured HF events on a Special Event Reporting form (refer to the Supplemental Appendix). Urgent visits were not limited to intravenous (IV) therapy. If treatment for HF was initiated or intensified, the investigator was asked to indicate whether this was initiation/significant dose augmentation of an oral diuretic, initiation of an IV diuretic, initiation of an IV vasoactive agent, mechanical circulatory support or surgical intervention, or mechanical fluid removal. For the present analyses, the main outcome was a composite of reported HF events defined as time to first new onset or worsening of HF leading to unscheduled hospital admission or an urgent clinic/office/emergency department visit (with initiation of or intensified diuretic/vasoactive therapy) or adjudicated CV death (including undetermined cause of death). Additional analyses examined time to first HF hospitalization or urgent visit requiring IV therapy (defined as IV diuretic or IV vasoactive therapy), time to first HF event alone, and CV death alone.

All analyses were performed in the entire population and in subgroups defined by presence of HF at baseline and HF subtype (with heart failure with preserved ejection fraction [HFpEF] and heart failure with reduced ejection fraction [HFrEF], as stated by the investigator), EF (<40%, 40%-<50% [denoting a mildly reduced EF] and ≥50%), body mass index, history of atherosclerotic cardiovascular disease (ASCVD), CKD (based on eGFR and UACR criteria of Kidney Disease: Improving Global Outcomes [KDIGO]),[16] and other relevant clinical variables. Serious adverse events and adverse events leading to discontinuation of study drug were summarized by treatment across baseline HF categories. All analyses were prespecified (refer to the Supplemental Appendix).

STATISTICAL ANALYSIS. The FLOW trial was event driven and designed to provide 90% power to detect a 20% relative risk reduction for semaglutide vs placebo for the primary outcome.[12,15] Assuming an event rate for the primary outcome of 7.5% per year in the placebo group, a minimum of 3,508 participants were to be enrolled in the trial.

Efficacy analyses were based on the intention-to-treat principle and were to include all unique participants who underwent randomization irrespective of adherence to semaglutide or placebo or changes to background medications. Data from participants who withdrew from the trial, died from causes not included in the main composite outcome, or were lost to follow-up were censored at the time of withdrawal, death, or last contact with the investigator. Time-to-event outcomes were analyzed using a stratified Cox proportional hazards model with treatment (semaglutide or placebo) and SGLT2 inhibitor use at baseline as the strata.

Subgroup analyses were performed by adding the interaction term between subgroup at baseline and treatment. Supplementary analyses were made when the underlying variable for the subgroup was quantitative (eg, body mass index), by including as a continuous variable and the interaction with treatment to evaluate whether a potential heterogeneity between treatment followed a linear trend. Likewise, time-to-event outcomes were plotted by treatment and HF subgroup at baseline using the Aalen-Johansen estimator and presented as cumulative incidences considering non-CV death or all-cause death as a competing event. The Ghosh-Lin regression model was used for analyses of time to first and recurrent event (total events) with treatment as a covariate. Absolute risk differences and corresponding numbers needed to treat (NNTs) were estimated using a generalized linear regression model with identity link function on the pseudo-observations from the Aalen-Johansen estimate at 2 and 3 years, respectively. Continuous outcomes were assessed at week 104 by analysis of covariance using multiple imputation for missing values under a missing-at-random assumption. Each imputed data set (n = 500) was analyzed using analysis of covariance with treatment and the baseline outcome as covariates and combined using Rubins method. Interaction P values were derived from an F-test of equality between the treatment differences across the relevant subgroups.

RESULTS

BASELINE DEMOGRAPHICS AND DISEASE CHARACTERISTICS. Overall, 3,533 participants were randomized to semaglutide (n = 1,767) or matching

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TABLE 1 Baseline Demographics and Clinical Characteristics According to HF Status at Baseline (Full Analysis Set)

Characteristics	HF at Baseline[a]	
	Yes (n = 678)	No (n = 2,854)
HF subtype		
HFpEF	325 (47.9)	–
HFrEF	123 (18.1)	–
Unknown	230 (33.9)	–
Sex		
Female	247 (36.4)	822 (28.8)
Male	431 (63.6)	2,032 (71.2)
Age, y	68.0 (63.0-73.0)	67.0 (61.0-73.0)
Ethnicity		
Hispanic or Latino	70 (10.3)	486 (17.0)
Other	590 (87.0)	2,241 (78.5)
Not reported	18 (2.7)	127 (4.4)
Race		
American Indian or Alaska Native	4 (0.6)	21 (0.7)
Asian	41 (6.0)	805 (28.2)
Black or African American	24 (3.5)	136 (4.8)
Native Hawaiian or other Pacific Islander	0 (0.0)	5 (0.2)
Not reported	7 (1.0)	72 (2.5)
Other	8 (1.2)	87 (3.0)
White	594 (87.6)	1,728 (60.5)
Body weight, kg	95.2 (83.0-108.8)	85.7 (73.4-100.3)
BMI, kg/m^2	33.3 (29.8-38.0)	30.8 (27.0-34.7)
HbA_{1c}, %	7.9 (7.1-8.8)	7.5 (6.8-8.5)
hsCRP, μg/L	3.4 (1.6-7.9)	2.3 (1.0-5.5)
eGFR, mL/min/1.73 m^2	45.0 (36.0-56.0)	45.0 (35.0-57.0)
eGFR group		
≥60 mL/min/1.73 m^2	118 (17.4)	601 (21.1)
<60 mL/min/1.73 m^2	560 (82.6)	2,252 (78.9)
UACR, mg/g	567.9 (212.9-1,244.7)	567.7 (234.4-1,320.0)
UACR group		
<30 mg/g	38 (5.6)	71 (2.5)
≥30-<300 mg/g	177 (26.1)	827 (29.0)
≥300 mg/g	463 (68.3)	1,955 (68.5)
NYHA functional class		
I	216 (31.9)	–
II	389 (57.4)	–
III	72 (10.6)	–
Missing	1 (0.1)	–
LVEF group		
<40%	69 (10.2)	14 (0.5)
40%-<50%	66 (9.7)	38 (1.3)
≥50%	308 (45.4)	757 (26.5)
Missing	235 (34.7)	2,045 (71.7)
Total cholesterol, mmol/L	4.3 (3.6-5.1)	4.2 (3.5-5.0)
HDL cholesterol, mmol/L	1.0 (0.9-1.2)	1.1 (0.9-1.3)
LDL cholesterol, mmol/L	2.2 (1.7-2.9)	2.1 (1.6-2.8)
Systolic blood pressure, mm Hg	138.0 (129.0-147.0)	138.0 (128.0-148.0)
Diastolic blood pressure, mm Hg	78.0 (71.0-84.0)	77.0 (70.0-83.0)

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placebo (n = 1,766) and were followed for a median of 3.4 years. At baseline, HF was reported for 342 participants (19.4%) randomized to semaglutide and 336 (19.0%) of those randomized to placebo. Among participants with HF, 31.9% (216 of 678) were categorized as NYHA functional class I, and the remainder were categorized as NYHA functional class II or III. Most of the participants with HF (47.9%; 325 of 678) were reported as having HFpEF, whereas 18.1% (123 of 678) were reported as having HFrEF (**Table 1**). HF subtype was unknown/unclassified in 33.9% of participants (230 of 678) with HF. Information on LVEF was recorded in 65.3% of participants (443 of 678) with HF and in 28.3% (809 of 2,854) of those without. Among those with HF, 45.4% (308 of 678) had an LVEF ≥50%, consistent with the main clinical diagnosis of HFpEF in this population.

Participants with HF at baseline, compared with those without, were more likely to be female (36.4% [247 of 678] vs 28.8% [822 of 2,854]), White (87.6% [594 of 678] vs 60.5% [1,728 of 2,854]) and of Other ethnicity (87.0% [590 of 678] vs 78.5% [2,241 of 2,854]), have a higher median body mass index (33.3 vs 30.8 kg/m^2) and glycated hemoglobin (7.9% vs 7.5%), and have a history of ASCVD (prior myocardial infarction or stroke; 39.6% [266 of 671] vs 19.2% [542 of 2,825]) and atrial fibrillation (18.4% [125 of 678] vs 5.0% [142 of 2,854]) (**Table 1**). Median eGFR and UACR did not differ between HF subgroups (45.0 vs 45.0 mL/min/1.73 m^2 and 567.9 vs 567.7 mg/g for participants with and without HF, respectively). A higher proportion of participants with HF at baseline were receiving beta-blockers, loop diuretics, and MRAs compared with those without HF; use of angiotensin-converting enzyme inhibitors or angiotensin II receptor blockers did not differ between groups (**Table 1**). In addition, participants with HF at baseline were more likely to be receiving insulin for treatment of their diabetes and were less likely to be receiving dipeptidyl peptidase-4 inhibitors or SGLT2 inhibitors. According to the KDIGO risk calculator, 68% of participants were at very high risk of kidney disease progression, kidney failure, CV events, or death. Baseline demographics and clinical characteristics according to HF subgroup at baseline are presented in Supplemental Table 1.

HF OUTCOMES IN THE OVERALL TRIAL POPULATION. In the overall population, the main composite HF outcome of HF events or CV death occurred less frequently in the semaglutide group (222 first events, rate 3.8 per 100 patient-years [PY] of follow-up) compared with placebo (292 first events, rate 5.1 per 100 PY of follow-up), resulting in a 27% lower relative risk in semaglutide-treated participants (HR: 0.73; 95% CI: 0.62-0.87; $P = 0.0005$) (**Figure 1A**). HF events alone (excluding CV death) were also lower (133 first events, rate 2.3 per 100 PY of follow-up) compared

with placebo (175 first events, rate 3.0 per 100 PY of follow-up), resulting in a 27% lower relative risk in semaglutide-treated participants (HR: 0.73; 95% CI: 0.58-0.92; $P = 0.0068$) (Figure 1B). The risk reduction for HF hospitalization or urgent visit requiring IV therapy (defined as IV diuretic or IV vasoactive therapy) was 26% in semaglutide-treated participants (HR: 0.74; 95% CI: 0.58-0.94; $P = 0.0154$) (Figure 1E). For those treated with semaglutide, the number of events were lower (117 first events, rate 2.0 per 100 PY of follow-up) than those treated with placebo (152 first events, rate 2.6 per 100 PY). Additionally, a risk reduction of 24% for HF hospitalization or urgent visit requiring IV diuretic therapy was reported in semaglutide-treated participants (HR: 0.76; 95% CI: 0.59-0.97; $P = 0.0248$). For this outcome, the number of events for the semaglutide group was 117 first events (rate 2.0 per 100 PY) and the corresponding event rate for the placebo group was 149 first events (rate 2.6 per 100 PY).

The risk reduction for CV death alone was 29% in semaglutide-treated participants (HR: 0.71; 95% CI: 0.56-0.89; $P = 0.0036$), with 123 first events in the semaglutide group (rate 2.1 per 100 PY of follow-up) and 169 first events in the placebo group (rate 2.9 per 100 PY of follow-up) (Supplemental Figure 1A).

Based on the absolute risk difference at 2 and 3 years, the NNTs to prevent 1 main composite outcome of HF or CV death were 34 (absolute risk difference: −0.03; 95% CI: −0.05 to −0.01) over 2 years and 24 (absolute risk difference: −0.04; 95% CI: −0.06 to −0.02) over 3 years. The corresponding NNTs to prevent 1 HF event were 50 (absolute risk difference: −0.02; 95% CI: −0.03 to −0.01) over 2 years and 38 (absolute risk difference: −0.03; 95% CI: −0.04 to −0.01) over 3 years.

In total, 750 events (including first and recurrent events) were reported for the main composite outcome of HF events or CV death, which compares with 514 when looking at first events only. The corresponding mean ratio for the semaglutide vs placebo group (mean ratio: 0.74; 95% CI: 0.60-0.90; event rates of 5.5 and 7.4 for semaglutide and placebo, respectively) did not differ substantially from that reported for first events alone. Similarly, the total burden of HF events (first and recurrent) was 458, compared with 308 for first events only (corresponding mean ratio: 0.76; 95% CI: 0.59-0.98; event rates of 3.4 and 4.5, respectively) (Supplemental Figure 2).

HF OUTCOMES ACCORDING TO HF STATUS AT BASELINE. Rates of HF events or CV death were higher (by ~2.5 fold) in participants with a HF compared with those without, but the relative risk reduction with semaglutide was similar in both groups. For the main composite outcome of HF events or CV death in participants with HF at baseline, there were 82 first events with a rate of 8.0 per 100 PY of follow-up in the semaglutide group compared with 101 first events with a rate of 10.9 per 100 PY of follow-up in the placebo group, resulting in a 27% lower relative risk (HR: 0.73; 95% CI: 0.54-0.98; $P = 0.0338$) (Figure 1C). Among participants without HF at baseline, there were 140 first events with a rate of 3.0 per 100 PY of follow-up in the semaglutide group compared with 191 first events with a rate of 4.1 per 100 PY of follow-up in the placebo group, resulting in a 28% lower relative risk (HR: 0.72; 95% CI: 0.58-0.89; $P = 0.0028$) (Figure 1C). Similarly, HF events alone were reduced following treatment with semaglutide in participants with and without HF,

TABLE 1 Continued

Characteristics	HF at Baseline[a]	
	Yes (n = 678)	No (n = 2,854)
Heart rate, beats/min	71.0 (64.0-78.0)	73.0 (65.0-81.0)
Concomitant diseases		
History of CV event		
No prior MI or stroke	405 (60.4)	2,283 (80.8)
Prior MI or stroke	266 (39.6)	542 (19.2)
Coronary heart disease	506 (74.6)	825 (28.9)
Atrial fibrillation	125 (18.4)	142 (5.0)
Hypertension	671 (99.0)	2,789 (97.7)
Coronary revascularization	210 (31.0)	481 (16.9)
PCI	130 (19.2)	299 (10.5)
CABG	100 (14.7)	204 (7.1)
Duration of diabetes, y	16.8 (10.9-22.1)	16.1 (10.7-22.6)
Concomitant medications		
Beta-blockers	519 (76.5)	1,324 (46.4)
Diuretics	481 (70.9)	1,299 (45.5)
Thiazides	75 (11.1)	538 (18.9)
Loop diuretics	321 (47.3)	569 (19.9)
MRA	113 (16.7)	144 (5.0)
ACE inhibitor/ARB	644 (95.0)	2,719 (95.3)
ARN inhibitor	2 (0.3)	1 (0.0)
Metformin	336 (49.6)	1,496 (52.4)
Sulfonylureas	173 (25.5)	718 (25.2)
SGLT2 inhibitor	82 (12.1)	468 (16.4)
DPP-4 inhibitor	123 (18.1)	782 (27.4)
Insulin	458 (67.6)	1,709 (59.9)

Values are n (%) or median (Q1-Q3). [a]1 participant had no HF status recorded.

ACE = angiotensin-converting enzyme; ARB = angiotensin II receptor blocker; ARN = angiotensin receptor-neprilysin; BMI = body mass index; CABG = coronary artery bypass graft; CV = cardiovascular; DPP-4 = dipeptidyl peptidase 4; eGFR = estimated glomerular filtration rate; HbA_{1c} = glycated hemoglobin; HDL = high-density lipoprotein; HF = heart failure; HFpEF = heart failure with preserved ejection fraction; HFrEF = heart failure with reduced ejection fraction; hsCRP = high-sensitivity C-reactive protein; LDL = low-density lipoprotein; LVEF = left ventricular ejection fraction; MI = myocardial infarction; MRA = mineralocorticoid receptor antagonist; PCI = percutaneous coronary intervention; SGLT2 = sodium-glucose cotransporter 2; UACR = urine albumin-to-creatinine ratio.

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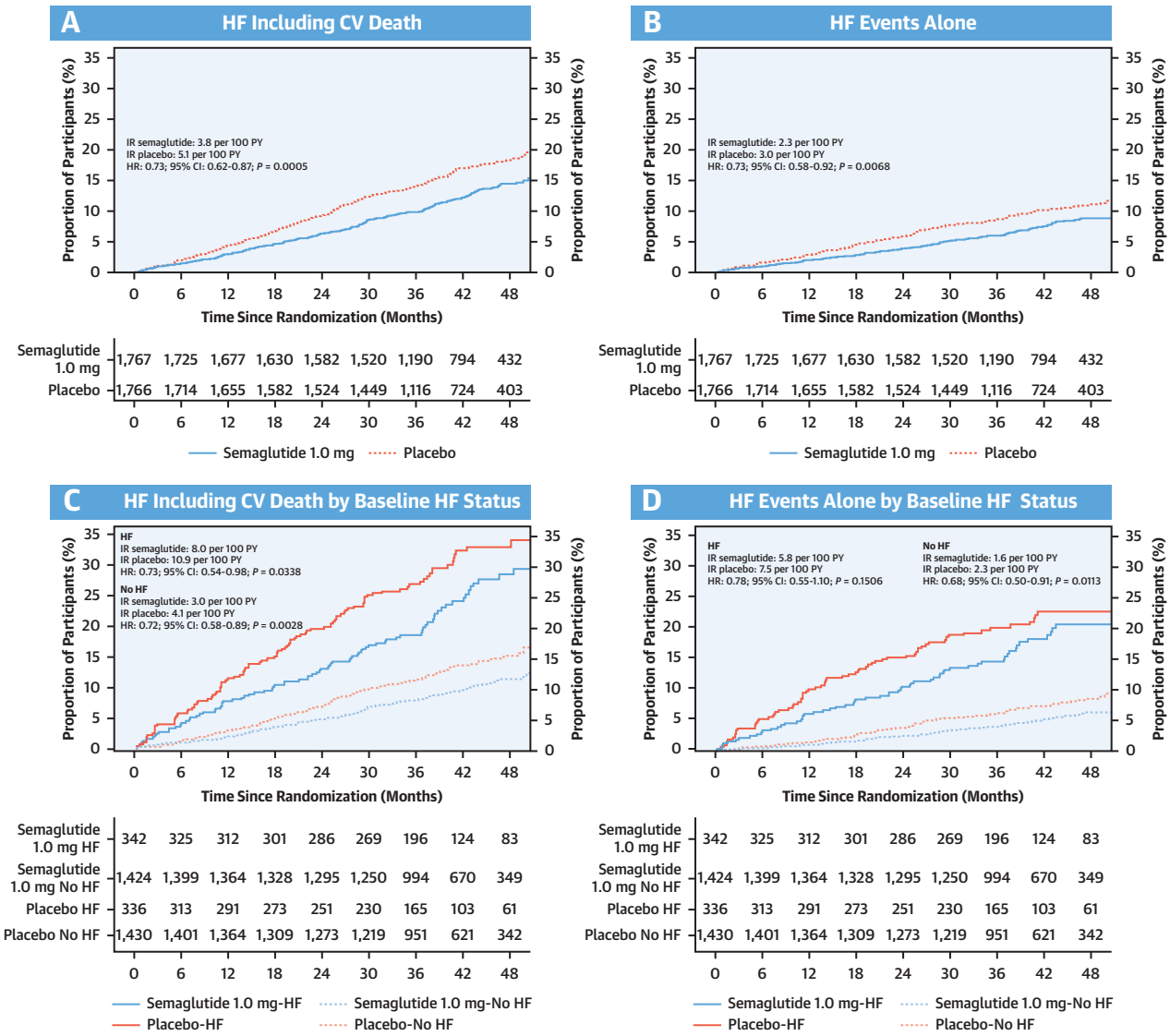
FIGURE 1 Cumulative Incidence Plots Showing the Time From Randomization to the First HF Event



Cumulative incidence plots showing the time from randomization to the first heart failure (HF) event by treatment for (A) composite HF events including cardiovascular (CV) death, (B) HF events, (C) composite HF events including CV death by HF status at baseline, (D) HF events by HF status at baseline, (E) HF hospitalization or urgent visit requiring intravenous (IV) therapy, and (F) HF hospitalization or urgent visit requiring IV therapy by HF status at baseline. In-trial, full analysis set. The cumulative incidence rate (IR) is calculated using the Aalen-Johansen method with non-CV death or all-cause death as a competing risk. IR expressed as events per 100 person-years (PY) of observations. IV therapy was defined as IV diuretic or IV vasoactive therapy.

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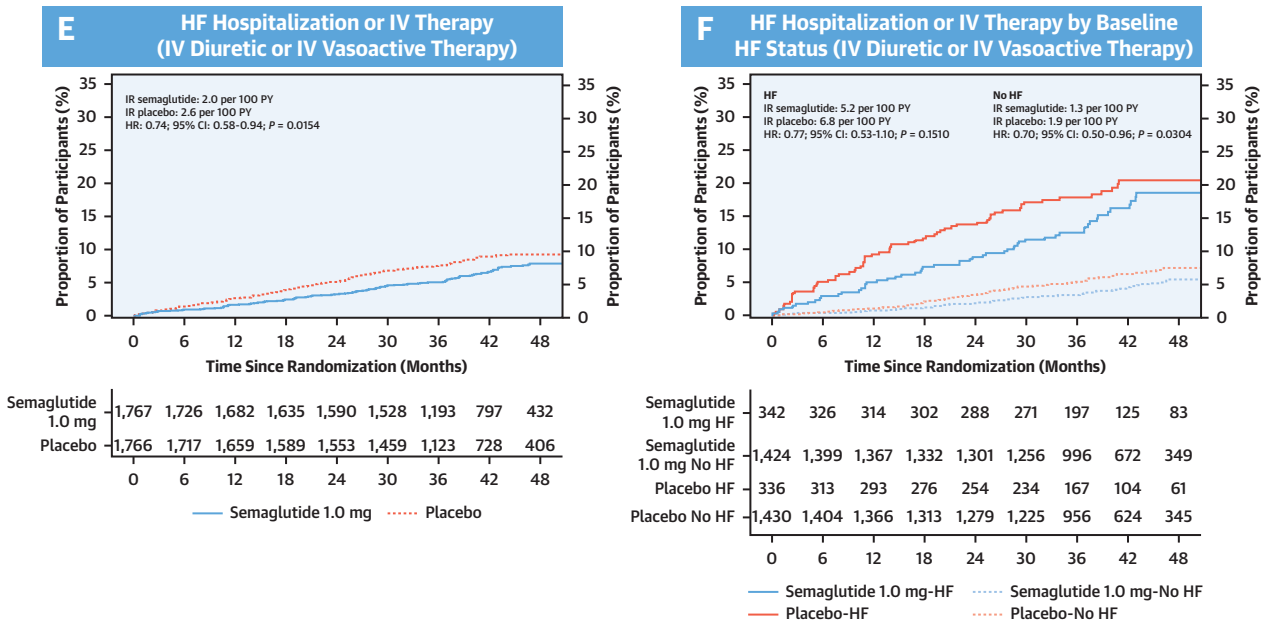
resulting in a 22% numerically lower relative risk (HR: 0.78; 95% CI: 0.55-1.10; $P = 0.1506$) (Figure 1D) in those with HF and a 32% lower relative risk (HR: 0.68; 95% CI: 0.50-0.91; $P = 0.0113$) (Figure 1D) in those without HF. In participants with HF at baseline who were treated with semaglutide, the risk reduction for HF hospitalization or urgent visit requiring IV therapy was 23% (HR: 0.77; 95% CI: 0.53-1.10; $P = 0.1510$) (Figure 1F). A risk reduction of 30% for the same outcome measure was reported in participants

FIGURE 1 Continued



without HF at baseline (HR: 0.70; 95% CI: 0.50-0.96; $P = 0.0304$) (Figure 1F).

For participants with HF at baseline, the risk reduction for CV death alone was 30% in semaglutide-treated participants (HR: 0.70; 95% CI: 0.46-1.05; $P = 0.0872$), with 40 events in the semaglutide group (rate 3.7 per 100 PY of follow-up) and 53 events in the placebo group (rate 5.1 per 100 PY of follow-up) (Supplemental Figure 1B). For participants without HF at baseline, the risk reduction for CV death was 29% in the semaglutide group (HR: 0.71; 95% CI: 0.53-0.93; $P = 0.0154$), with 83 events in the semaglutide group (rate 1.7 per 100 PY of follow-up) and 116 events in the placebo group (rate 2.5 per 100 PY of follow-up).

Based on the absolute risk difference at 2 and 3 years, the NNTs to prevent 1 main composite outcome of HF or CV death in participants with HF at baseline were 15 over 2 years and 12 over 3 years, with corresponding absolute risk differences (semaglutide vs placebo) of −0.07 (95% CI: −0.12 to −0.01) and −0.08 (95% CI: −0.15 to −0.02), respectively. The corresponding NNTs to prevent 1 HF event in participants with HF at baseline were 21 over 2 years and 18 over 3 years, with absolute risk differences of −0.05 (95% CI: −0.10 to 0.00) and −0.05 (95% CI: −0.11 to 0.00). For participants without HF at baseline, the NNTs to prevent 1 main composite outcome of HF or CV death were 46 over 2 years and 32 over 3 years, with corresponding absolute risk differences of −0.02 (95% CI: −0.04 to −0.00) and −0.03 (95% CI: −0.05 to −0.01). The corresponding NNTs to prevent 1 HF event were 71 over 2 years and 49 over 3 years, with absolute risk differences of −0.01 (95% CI: −0.03 to −0.00) and −0.02 (95% CI: −0.04 to −0.01).

HF OUTCOMES IN CLINICALLY RELEVANT SUBGROUPS. The benefits of semaglutide on the main composite outcome of HF events or CV death, as well as HF events alone, were similar across predefined subgroups (Figure 2). Notably, approximately 15% of the enrolled participants in the overall population were treated with an SGLT2 inhibitor at baseline (Table 1). Although fewer participants with HF than without reported SGLT2 inhibitor use at baseline (12.1% [82 of 678] vs 16.4% [468 of 2,854], respectively), there was no significant difference between treatment and baseline SGLT2 inhibitor use with respect to the main composite HF outcome (P interaction = 0.7281) (Figure 2A). MRA use was infrequent at baseline, but higher in those with HF than in those without (Table 1). The effects of semaglutide on HF or CV death and on HF events alone were similar in those with and without a history of MRA use. In addition, the benefit of semaglutide on HF outcomes was generally similar across CKD subgroups defined by eGFR, UACR, or KDIGO risk class (Figure 2). Notably,

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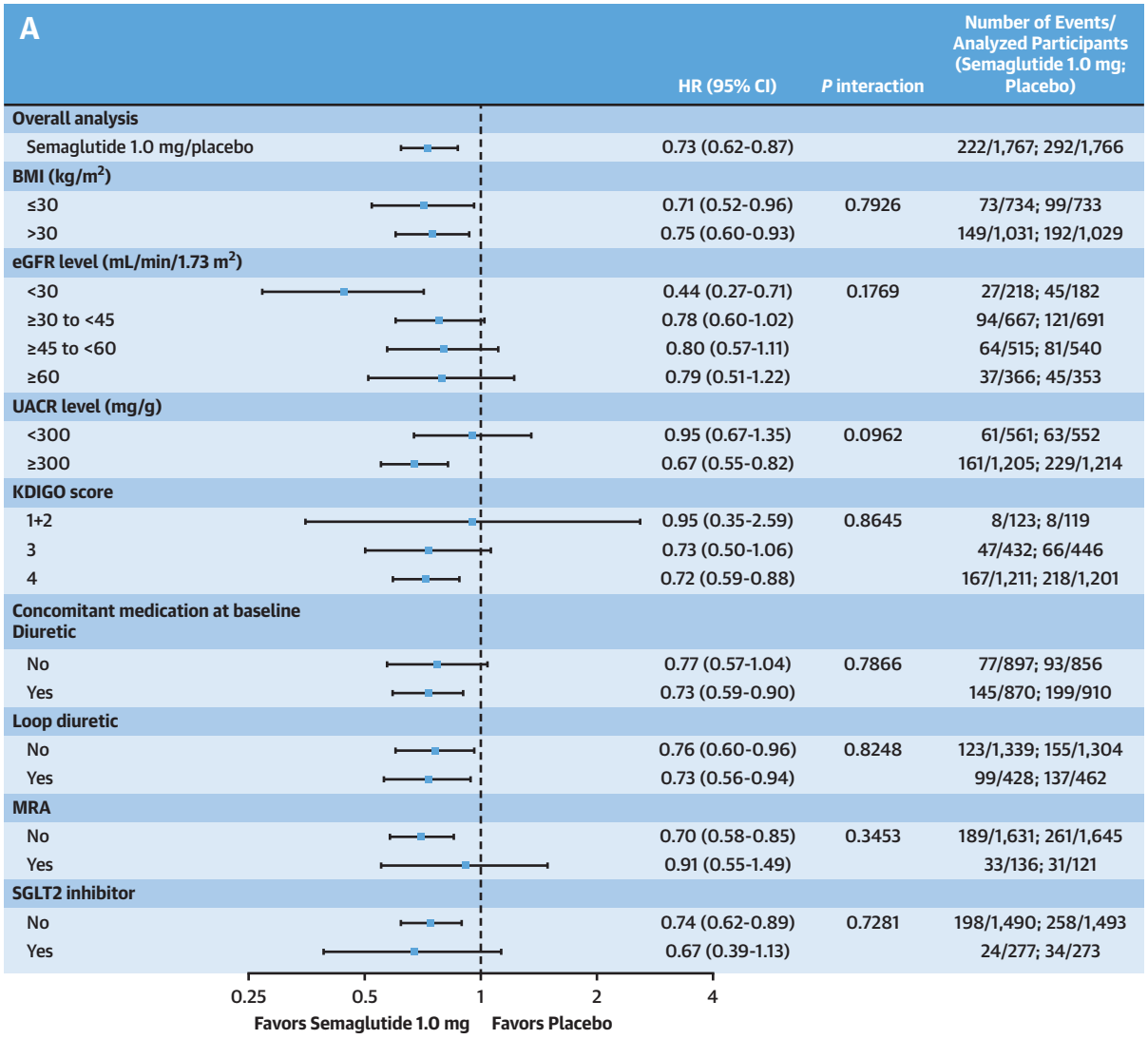
FIGURE 2 Time From Randomization to First Events in Prespecified Subgroups



Time from randomization to (A) first composite HF events or CV death and (B) first HF events in prespecified subgroups. In-trial, full analysis set. Analysis is based on a Cox proportional hazards model with treatment and subgroup as fixed factors and stratified by sodium-glucose cotransporter 2 (SGLT2) inhibitor use at baseline. For analyses with continuous covariate, estimated HRs and corresponding CIs were calculated in a Cox proportional hazards model with interaction between treatment and the covariate factor. For time to first composite HF events or CV death in subgroups based on body mass index (BMI), estimated glomerular filtration rate (eGFR), and urine albumin-to-creatinine ratio (UACR), the corresponding trend test interaction *P* values were 0.6315, 0.1989, and 0.2112, respectively. For time to first HF events, the corresponding trend test interaction *P* values were 0.5864, 0.0251, and 0.7969, respectively. KDIGO = Kidney Disease: Improving Global Outcomes; MRA = mineralocorticoid receptor antagonist; other abbreviations as in Figure 1.

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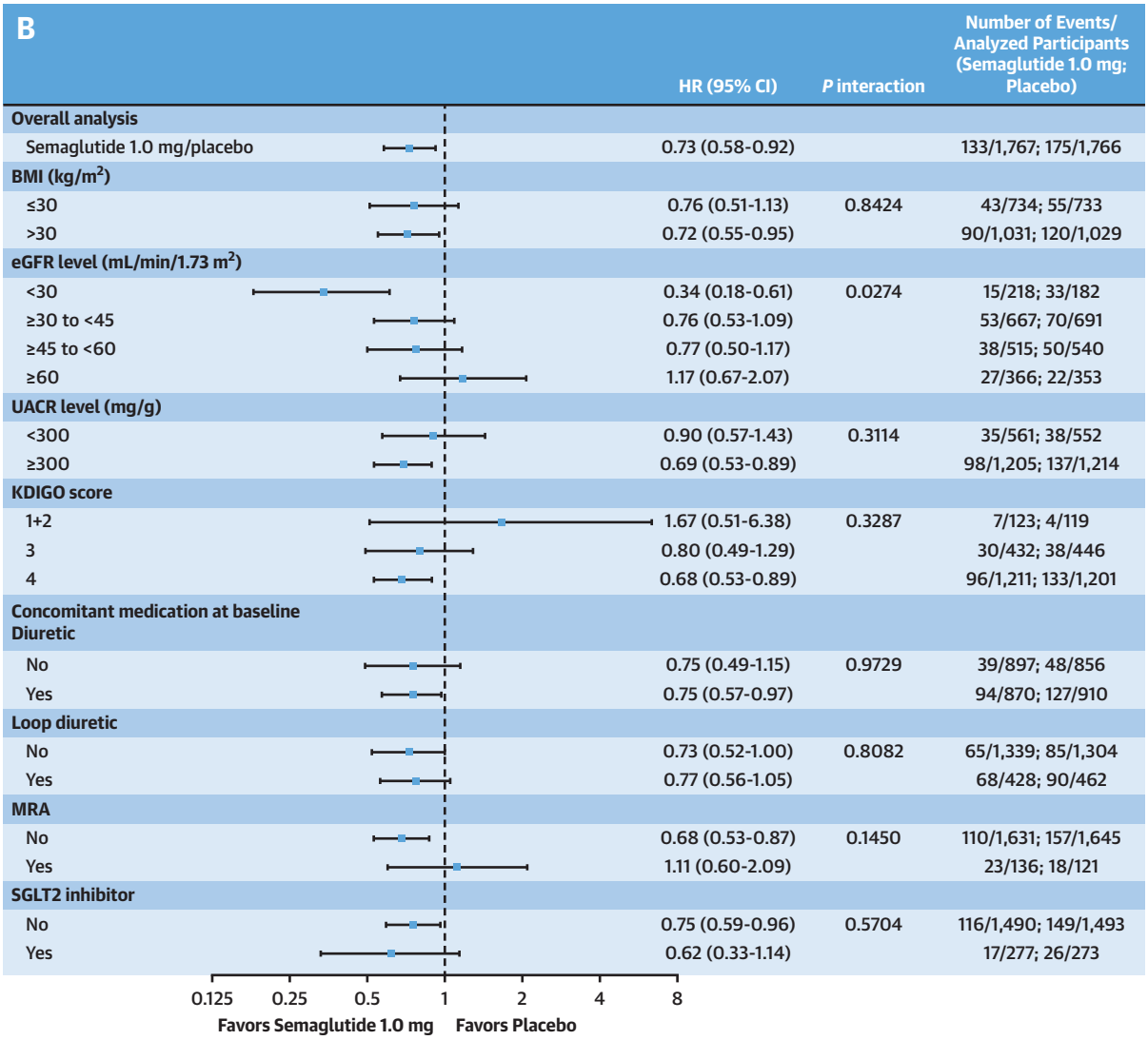
semaglutide significantly reduced the risk of HF events in participants with lower eGFR (*P* interaction = 0.0274) (Figure 2B).

There was no indication of heterogeneity in the effect of semaglutide on HF across HF subtypes, NYHA functional class or LVEF subgroups at baseline (Figure 3). The risk of HF outcomes, regardless of treatment, was generally higher in participants categorized as NYHA functional class III and those with the HFrEF subtype.

A history of a CV event (prior myocardial infarction or stroke) was more common among participants with HF at baseline than those without (Table 1) but did not affect the relative risk reduction for HF events or CV

FIGURE 2 Continued



death (HR: 0.64; 95% CI: 0.52-0.80 [rates of 3.1 per 100 PY of follow-up and 4.8 per 100 PY of follow-up for semaglutide and placebo groups, respectively], and HR: 0.95; 95% CI: 0.70-1.29 [rates of 6.5 per 100 PY and 6.8 per 100 PY] for participants with a history of ASCVD and those without, respectively). Similarly, the relative risk reduction for HF events was not affected by history of ASCVD (HR: 0.63; 95% CI: 0.47-0.84 [rates of 1.8 per 100 PY and 2.8 per 100 PY for semaglutide and placebo groups, respectively] in participants with ASCVD, and HR: 0.93; 95% CI: 0.64-1.36 [rates of 4.1 per 100 PY and 4.5 per 100 PY] in participants without ASCVD).

Consistent with effects reported in the primary analysis,[12] semaglutide decreased body weight, systolic blood pressure, and glycated hemoglobin to a similar degree in those with and without HF at baseline (Supplemental Figure 3).

ADVERSE EVENTS. Serious adverse events (**Table 2**) were reported more frequently among participants with HF at baseline compared with those without but were less frequent for those in the semaglutide group compared with the placebo group (57.6% [197 of 342] vs 66.1% [222 of 336]) and without HF (47.8% [680 of 1,424] vs 50.9% [728 of 1,430]). Regardless of HF status at baseline, adverse events leading to permanent treatment discontinuation were more common in participants receiving semaglutide vs those receiving placebo (12.9% [44 of 342] vs 10.4% [35 of 336] in those with HF and 13.3% [189 of 1,424] vs

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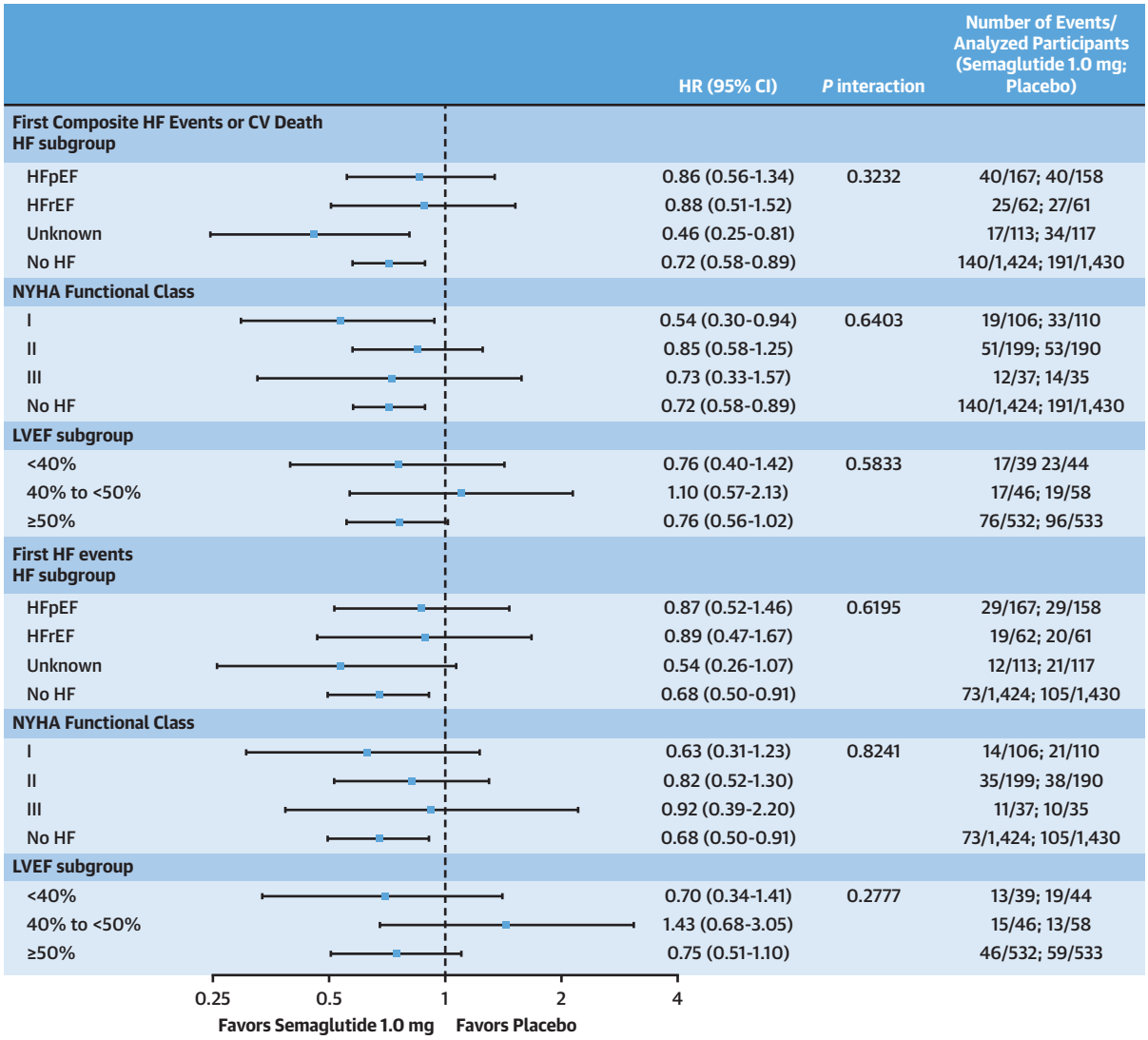
FIGURE 3 Time From Randomization to First Events or Death According to HF Subtype, NYHA Functional Class, and LVEF Category at Baseline



Time from randomization to first composite HF events or CV death and first HF events according to HF subtype, NYHA functional class, and left ventricular ejection fraction (LVEF) category at baseline. In-trial, full analysis set. Analysis is based on a Cox proportional hazards model with treatment and subgroup as fixed factors and stratified by SGLT2 inhibitor use at baseline. HFpEF = heart failure with permanent ejection fraction; HFrEF = heart failure with reduced ejection fraction; other abbreviations as in Figures 1 and 2.

12.3% [176 of 1,430] in those without HF at baseline), driven mainly by discontinuations caused by gastrointestinal disorders (3.2% [11 of 342] vs 0.9% [3 of 336] in those with HF and 4.8% [68 of 1,424] vs 1.2% [17 of 1,430] in those without HF at baseline) (Table 2).

DISCUSSION

This prespecified analysis of the FLOW trial demonstrated that semaglutide decreased a composite outcome of time to first HF events or CV death by 27%, HF events alone by 27%, and CV death by 29% compared with placebo in participants with CKD and T2D. Although the absolute event rates were higher among those participants with a history of HF than those without, the relative risk reductions for the composite of HF or CV death and for HF events alone were similar in those with or without HF at baseline and in participants with HFpEF or HFrEF. The effects of semaglutide were also comparable across a variety of clinically relevant subgroups, including those defined by KDIGO risk class, prior ASCVD events, and

background medications, including SGLT2 inhibitors and MRAs. In addition to other kidney and CV benefits, these data demonstrate that semaglutide decreases clinically meaningful HF outcomes in a high-risk population with T2D and CKD regardless of HF at baseline.

Most CV outcomes trials of GLP-1 RAs in T2D have demonstrated numeric reductions in time to first HF events.[17] However, only the Harmony Outcomes study, which tested albiglutide compared with placebo in participants with T2D and CVD (20% with HF at baseline), and the AMPLITUDE O (Effect of Efpeglenatide on Cardiovascular Outcomes) trial, which tested efpeglenatide vs placebo, also in participants with T2D and CVD (18% with HF at baseline), demonstrated significant reductions in HF hospitalization rates.[18,19] The CREDENCE study, which examined the efficacy of the SGLT2 inhibitor canagliflozin in participants with T2D and CKD (15% with HF at baseline), found a significant reduction in the rate of hospitalization for HF in those receiving active treatment.[10] Similarly, the DAPA-CKD (Dapagliflozin and Prevention of Adverse Outcomes in Chronic Kidney Disease) study (29% with HF at baseline) recorded a reduction in a composite of HF hospitalization or CV death.[20]

In FLOW, most participants with HF had a history of HFpEF or HF with mid-range EF, and not HFrEF. This is typical for both the population with T2D, who are often overweight or obese, and those with CKD. FLOW may represent a more diseased population than in previous CKD trials, including DAPA-CKD, which may account, in part, for the higher annualized risk of CV death in the placebo group observed herein. Two recent 52-week studies assessed once-weekly semaglutide 2.4 mg in participants with obesity-related HFpEF both with and without T2D (STEP-HFpEF [Research Study To Investigate How Well Semaglutide Works in People Living With Heart Failure and Obesity] and STEP-HFpEF DM [Semaglutide Treatment Effect in People With Obesity and Heart Failure With Preserved Ejection Fraction and Diabetes Mellitus] trials).[13,21] These trials demonstrated significant improvements in HF symptoms and physical limitations, assessed by a commonly used health status instrument (Kansas City Cardiomyopathy Questionnaire Clinical Summary Score). The 6-minute walk distance was also improved, as was a composite outcome (all-cause death, HF events, Kansas City Cardiomyopathy Questionnaire Clinical Summary Score, and 6-minute walk distance). In a pooled analysis of the 2 trials, significant reductions were observed in HF hospitalization or urgent visits in which IV therapy was administered and the composite of CV death or HF events.[22] Thus, the effects of once-weekly subcutaneous semaglutide 1.0 mg in participants with T2D and CKD on HF outcomes in the FLOW trial are broadly consistent with the results of these 2 dedicated HF studies using a higher dose of semaglutide, but also extend the benefits to a population at high risk with no history of HF. It is important to note, however, that because FLOW was not a dedicated HF trial, the patient population (even among those participants with a history of HF) was quite different from that of the STEP-HFpEF program.

TABLE 2 Summary of Adverse Events According to HF Status at Baseline (Full Analysis Set)

Adverse Event	HF at Baseline		No HF at Baseline	
	Placebo (n = 336)	Semaglutide 1.0 mg (n = 342)	Placebo (n = 1,430)	Semaglutide 1.0 mg (n = 1,424)
SAEs	222 (66.1)	197 (57.6)	728 (50.9)	680 (47.8)
GI disorders[a]	18 (5.4)	12 (3.5)	76 (5.3)	83 (5.8)
AEs leading to treatment discontinuation[b]	35 (10.4)	44 (12.9)	176 (12.3)	189 (13.3)
GI disorders[a]	3 (0.9)	11 (3.2)	17 (1.2)	68 (4.8)

Values are n (%). [a]System organ class. [b]All permanent treatment discontinuations until the end-of-treatment visit are included.

AE = adverse event; GI = gastrointestinal; HF = heart failure; SAE = serious adverse event.

The mechanisms by which semaglutide decreases HF outcomes in patients with T2D and CKD are not fully understood. It is highly unlikely that the observed benefits in HF can be ascribed to improvements in glycemia alone.[23-26] In the STEP-HFpEF trial, similar improvements in HF symptoms were observed in a population without T2D as were seen in STEP-HFpEF DM in a population with T2D.[13,21] It is possible that the weight loss observed with semaglutide underpins some of the improvements in HF. However, multiple lines of evidence indicate that there are additional, direct disease-modifying effects of semaglutide on the pathobiology of HF, which are likely independent of weight loss. In the STEP-HFpEF program, the HF benefits of semaglutide vs placebo were similar in participants with and without T2D, despite approximately 40% more weight loss in nondiabetic than diabetic participants, −10.7% vs −6.4% placebo-adjusted, respectively.[13,21] Improvements in HF-related symptoms and physical limitations with semaglutide were also more pronounced in participants with more vs less severe HF at baseline (higher N-terminal pro-B-type natriuretic peptide, higher NYHA functional class, taking loop diuretics), although the degree of weight loss was similar across these subgroups. Furthermore, in the STEP-HFpEF program, semaglutide significantly lowered N-terminal pro-B-type natriuretic peptide

levels regardless of the magnitude of weight loss, which is suggestive of decongestive effects, and substantially reduced diuretic requirements.[13,21] Such effects may be more pronounced in patients with greater adiposity, and we note that the majority of participants in FLOW had overweight/obesity. GLP-1 has a natriuretic effect, which has been proposed as a potential mechanism for the HF benefits associated with GLP-1 RAs.[27] Measurement of natriuretic peptide levels was not performed as part of the study procedures in FLOW. Another possibility is that semaglutide decreases the proinflammatory milieu that is associated with HF. In STEP-HFpEF and STEP-HFpEF DM, treatment with semaglutide led to a significant decrease in high-sensitivity C-reactive protein.[12]

Strengths of the FLOW trial include the enrollment of a large population of participants with T2D and CKD who are at high risk for HF or with known HF, and the relatively long duration of follow-up. By including a diverse population, we were able to demonstrate that the benefits of semaglutide extended beyond those with prevalent HF (including those with HFrEF) and were also evident across a broad range of clinically relevant subgroups, including those without prevalent HF at baseline.

STUDY LIMITATIONS. We acknowledge that the low number of Black individuals included in the current analysis does not reflect the high burden of HF observed in the Black population more broadly. An important limitation is that HF events were not adjudicated, and instead, were captured via investigator report. However, possible HF events and completion of the investigator report were meticulously monitored during the trial. Prior studies have also shown that investigator-identified HF events are broadly comparable to adjudicated events.[28] Because this trial was initiated before SGLT2 inhibitors and nonsteroidal MRAs were approved to decrease the progression of CKD, a minority of participants were treated with these guideline-directed therapies, which may also have HF benefits. Other limitations include the fact that data regarding LVEF were not systematically collected, and because of the way the data were gathered, LVEF categorizations were <40%, 40% to <50%, and ≥50%, in line with the accepted definitions for HF at trial conception. The clinical phenotype of HF was unknown in approximately one-third of participants, and natriuretic peptide levels were not systematically collected.

CONCLUSIONS

This prespecified analysis of the FLOW trial demonstrated that once-weekly subcutaneous semaglutide 1.0 mg significantly reduced the risk of a composite outcome of HF events or CV death, and HF events alone, by 27% in a high-risk population with T2D and CKD. The risk of CV death alone was reduced by 29%. The beneficial effects of semaglutide were similar in participants with and without HF at baseline and across a range of clinically relevant subgroups over a 3.4-year mean follow-up.

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Data will be shared with bona fide researchers who submit a research proposal approved by the independent review board. Individual patient data will be shared in data sets in a deidentified and anonymized format. Data will be made available after research completion and approval of the product and product use in the European Union and the United States. Information about data access request proposals can be found on the Novo Nordisk Trials website.

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APPENDIX For supplemental figures, a table, and a form, please see the online version of this paper.